# Transdisciplinary Research Implementation: Integrated Teams & Hybrid Models

## For National Security Applications


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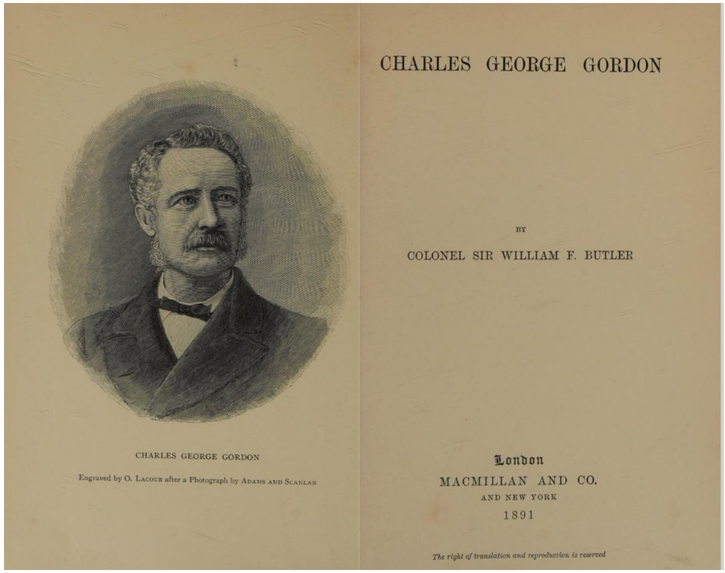
- Some Context
- Transdisciplinary Research Methods
- Transdisciplinary Methods & Ontologies
- Conclusions / Future Research Recommendations

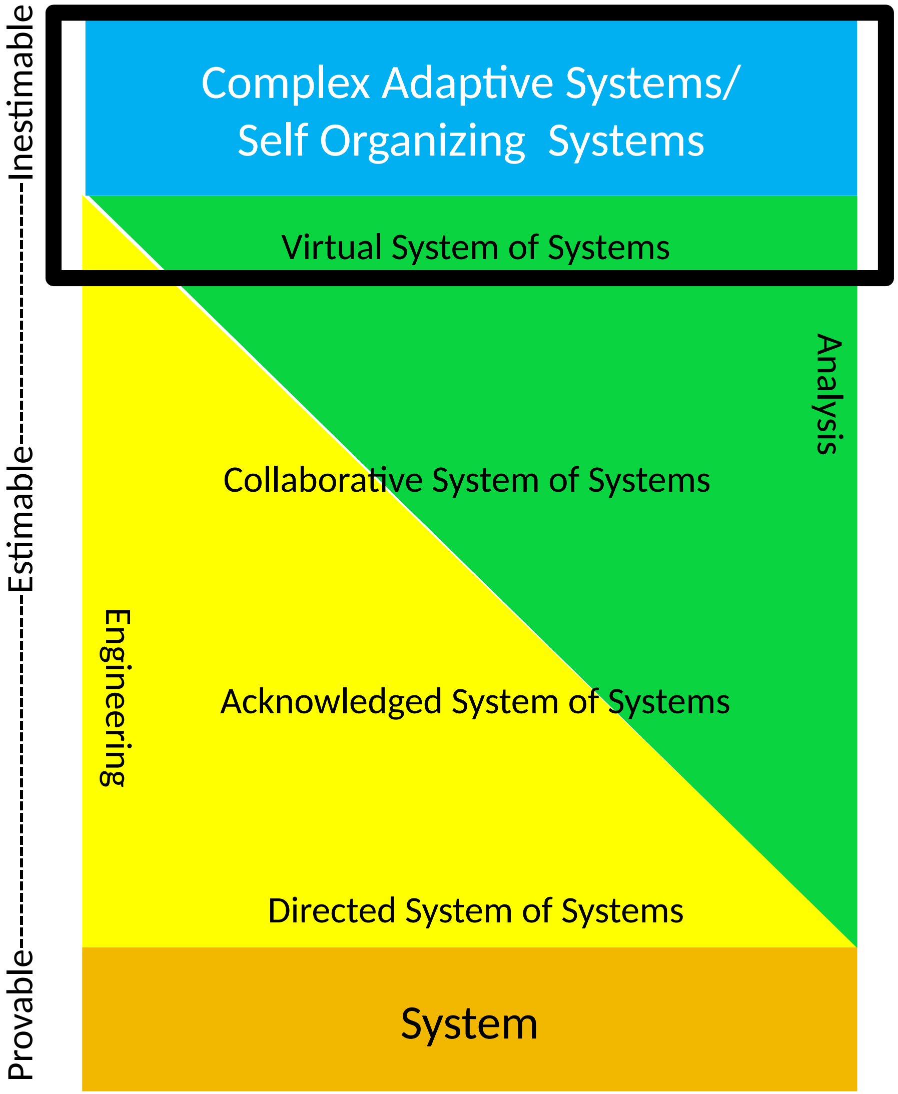
*"The nation that will insist on drawing a broad line of demarcation between the fighting man and the thinking man is liable to find its fighting done by fools and its thinking done by cowards."*

Sir William Francis Butler in Charles George Gordon, MacMillan and Co, 1891.



# The Current Research and Analytic Environment

(Through a **mostly** Engineering and OR Lens)
[Using **mostly** Engineering Vocabulary]



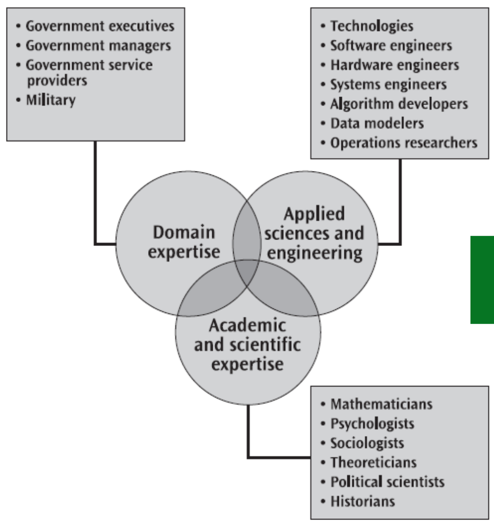

- National Security Activities all occur in a Complex Adaptive System
  - Internationally
  - Homeland
- Many (most) research, analysis, and engineering specialties are attempting to understand Complex Adaptive Systems:
  - Mostly independently
  - Through different lenses
  - Arriving at incomplete and potentially inconsistent conclusions
- **Through a decision maker's lens – Not Helpful!**
- **This is caused by a lack of shared understanding:**
  - Inconsistent and often conflicting goals / constraints
  - Limited Interoperability
    - Machine to Machine
    - Human to Machine
    - Human to Human
      - Skill to Skill
      - Organization to Organization
- Inability to effectively capitalize on the zetabytes of potentially useful data:
  - Can't sort the wheat from the chaff
  - Can't synthesize it
  - Can't reason over it

# Transdisciplinary Research – Next Generation OR (2 of 2)



Circa 2007

Figure 1. Venn Diagram of Model Team




State of the Art

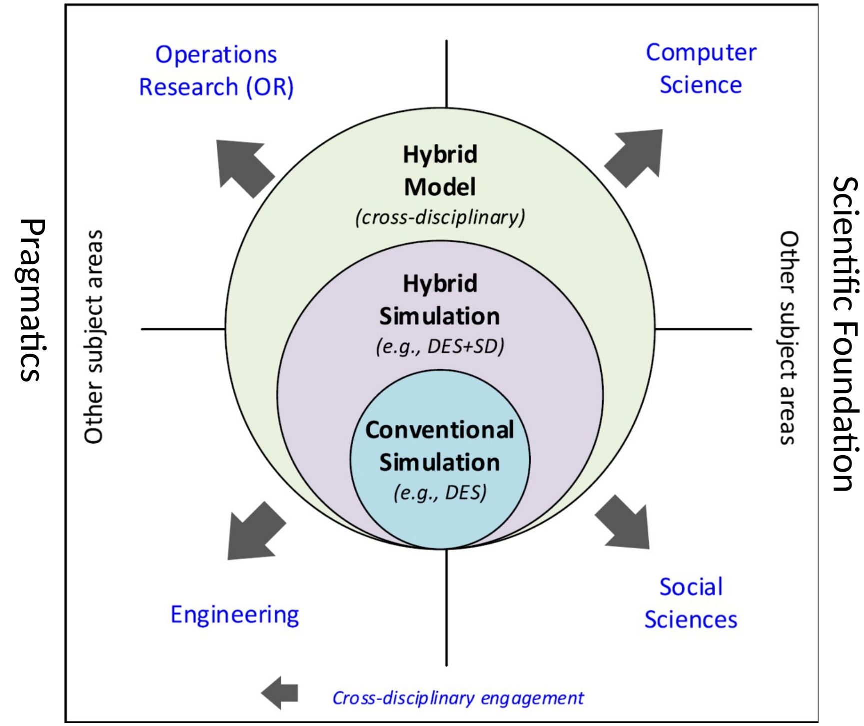
First Principles Endure

Table 2. Three Features of Good Model

| Feature | Explanation |
|---|---|
| Clearly bounded | A concept of interest or variable is clearly inside or outside the scope. |
| Logically consistent | The variables inside the model fit together and are clearly related to the relevant goals and decisions identified at the outset. |
| Penetrable | The team can explain to an executive how the model works and why it does what it does. |

From Maxwell, D. & Loerch A., [Executive's guide to practical computer models | IEEE Journals & Magazine | IEEE Xplore]

Circa 2025

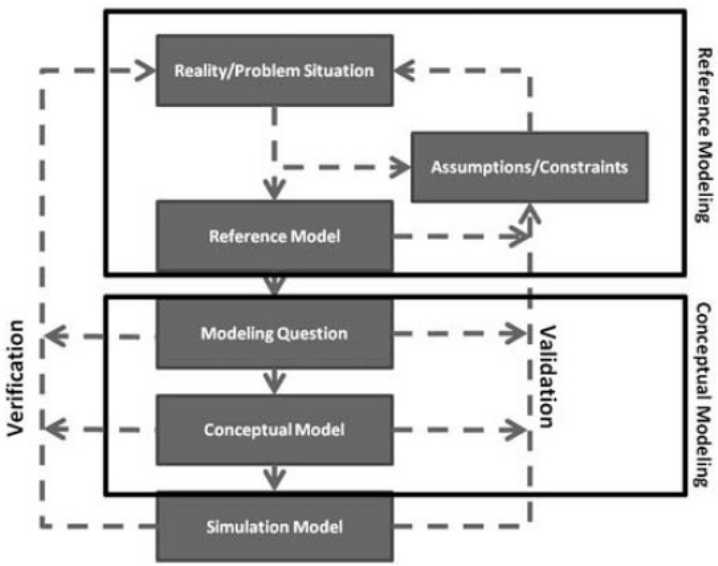



From Tolk, A. et al., [Hybrid models as transdisciplinary research enablers - ScienceDirect]

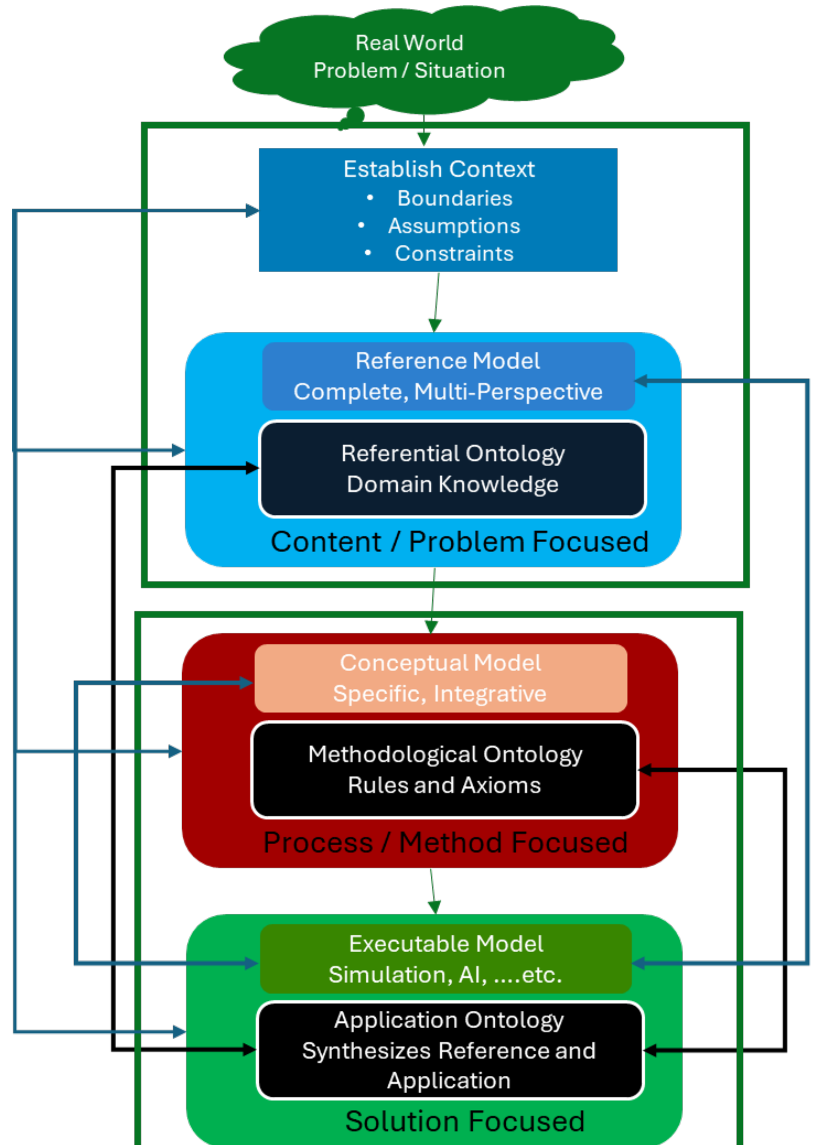
- Scientific Foundation
  - Multi-dimensional Clarity and Logical Consistency:
    - Geospatially
    - Temporally
    - Behaviorally
- Pragmatically
  - Responsiveness to intent (DSQ's, Competency Questions)
  - Debt Balance
    - Extendibility
    - Interoperability
    - Technical
  - Engineering Requirements
    - Computational Feasibility
    - Software Design Principles
      - Artifacts
      - **Security!!!**

# Hybrid Models and Ontologies






M&S System Development Framework
From Tolk et al. (2013)




Hybrid Modeling System Development Framework

# Sample Program Interoperability Comparisons



As-is (Sample System of Systems)

**Palantir – JADC2 Interoperability (Current State) Levels of Conceptual Interoperability Model (LCIM) Assessment**

| Interoperability Level | Score | Rationale |
|---|---|---|
| L6- Conceptual | X | Occurs outside current System of Systems framework |
| L5-Dynamic | X | No mechanism for change if context not shared |
| L4-Pragmatic | X | Context not shared with other systems |
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Current Gap

Could-be (Sample System of Systems)

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# Transdisciplinary Research Success Factors (1 of 2)



- A clearly bounded context is Required– You can't boil the ocean (Clearly Bounded)
- Scientific / Technical / Engineering Factors
  - Disciplined Software Engineering is essential:
    - Because:
      - The complexity of the supporting software will grow (Penetrable)
      - Security must be considered from the beginning (Pragmatic)
    - But:
      - Engineering shouldn't stymie creativity at the intersection of disciplines (Scientific Foundation)
      - Creativity must stay in the feasible region (Pragmatic)
    - There will be friction
  - An underlying ontology is essential (Clearly Bounded, Logically Consistent)
    - Facilitates shared understanding (Interoperability)
      - Limiting ambiguity, streamlining terminology
      - Enhances human communication and machine reasoning
    - But:
      - Avoid it just becoming a knowledge warehouse (Pragmatic)
      - Bound its use ( Scientific Foundation)

# Transdisciplinary Research Success Factors (2 of 2)



- Structural / Process Factors
  - An integrated approach to artifact management is essential
    - Software
    - Ontologies
    - Data
    - Documentation
- Social Factors
  - Team composition is critical
    - Members need to:
      - Know what they know
      - Know what they don't know
      - Listen, learn, and adapt
    - Leaders need to:
      - Know how to facilitate debate
        - » When to compromise and why
        - » When to hold the line on a key principle
      - Do it all politely, with genuine curiosity and respect for alternatives
      - Change the way you think about your home discipline in light of novel discoveries about partner disciplines

# Transdisciplinary Research Next Steps



- Refine and publish best practices for Transdisciplinary Research
  - Software Development
  - Ontology Development
  - Model Development (Includes Ontologies)
  - Data Management
- Adopt and IMPLEMENT open standards
- Explore the limits of in the context of bounded problems
- Conduct research into approaches to simplification of components
- What should **APPLIED** Ontologists do?
  - Communicate your value in operational terms
  - Recognize the limits of your models – and then work with experts in those disciplines. Key areas e.g.:
    - Decision Modeling
    - Dynamic Modeling (Simulation)
    - Uncertainty modeling

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- Beverley, J., G. De Colle, M. Jensen, C. Benson, and B. Smith. 2024. “Middle architecture criteria”. *arXiv preprint*
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- Beverley, J., Maxwell D. et al. (2024) **Ontologies, Arguments, and Large-Language Models,** In Ítalo Oliveira (Ed.), *Joint Ontologies Workshops (JOWO)*, CEUR-WS.org, Vol. 3882, pp. 1–9.
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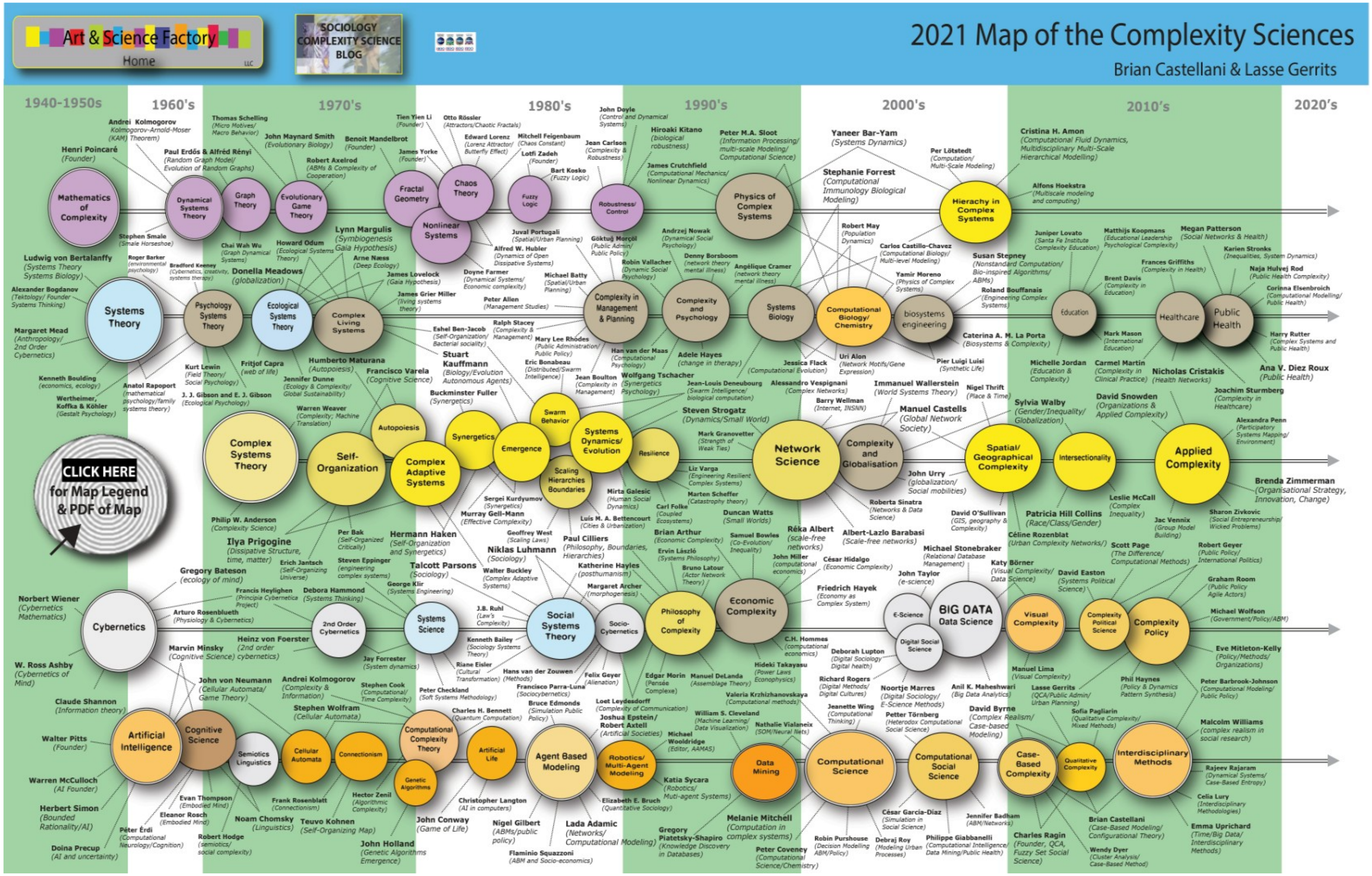
- Tolk, A., Diallo, S. Y., Padilla, J. J., & Herencia-Zapana, H. (2013). Reference modelling in support of M&S—foundations and applications. *Journal of Simulation*, *7*(2), 69–82. https://doi.org/10.1057/jos.2013.3




https://art-sciencefactory.com/complexity-map_feb09.html

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